TELESCOPING KELLER MAPS OF AFFINE THREE-SPACE: DEGREES AND EXPLICIT SYMMETRIC MONODROMY

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Abstract. We give a telescoping construction of polynomial self-maps of affine three-space with Jacobian determinant -2. Its exact polynomiality criterion consists of three scalar identities at $a = h(0) \neq 0$, and a member defined by a polynomial E of degree $n \geq 1$ has geometric degree exactly $n + 2$. Shifted Legendre polynomials supply members for every $n \geq 1$. An explicit degree-five member has primitive generic-fiber equation

$$\nu^5 - 5\nu^3 + 4\nu^2 - bc\nu + 2ac^2 = 0$$

and geometric monodromy S_5; hence its generic algebraic inverse is not solvable by radicals. We also record an S_4 member and the resulting degree spectrum $\mathbb{N} \setminus \{2\}$ in every dimension at least three. The exclusion of degree two is the already-known quadratic corollary of the classical Galois case of the Jacobian problem, not a new result here.

1. Scope and conventions

A *Keller map* is a polynomial map with nonzero constant Jacobian determinant. Its *geometric degree* is the degree of the induced extension of rational function fields, equivalently the cardinality of a general geometric fiber. We work over $\mathbb{C}$, although the displayed identities are defined over $\mathbb{Q}$.

The claims below are deliberately limited to the construction, its degree, and the two explicitly certified monodromy groups. Same-day public work also reported constructions in every generic fiber degree at least three [2]; we record that overlap neutrally and make no claim here about exact priority or novelty. The accompanying script performs exact symbolic and finite-field checks.

2. The telescoping construction

Put

$$s = x^2 z, \qquad t = xy, \qquad u = 1 + t.$$

For $E(\nu) \in \mathbb{C}[\nu]$, define primitives with zero constant term

$$A(\nu) = \int_0^\nu \xi E(\xi)\, d\xi, \qquad C(\nu) = 2\int_0^\nu E(\xi)\, d\xi.$$

Given $h(t) \in \mathbb{C}[t]$, set

$$g = h(t) - s, \qquad \nu = ug, \qquad S = A(ug) + ug^2, \qquad T = C(ug) + 2g.$$

Consider the initially rational expression

$$F_{E,h}(x, y, z) = \left(\frac{S(x^2z, xy)}{x^2 g(x^2z, xy)^2}, \frac{T(x^2z, xy)}{xg(x^2z, xy)}, xg(x^2z, xy) \right). \tag{1}$$

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Theorem 2.1 (Telescoping family)**.** *Let E have degree $n \geq 1$ and put $a = h(0)$. The expression (1) is a polynomial map if and only if $a \neq 0$ and*

$$A(a) = -a^2, \tag{2}$$

$$C(a) = -2a, \tag{3}$$

$$1 + E(a) + h'(0)\frac{E(a)+2}{a} = 0. \tag{4}$$

When these conditions hold,

$$\det \mathrm{Jac}(F_{E,h}) = -2, \qquad \deg_{\mathrm{geom}}(F_{E,h}) = n + 2.$$

Proof. First suppose that $a \neq 0$. Write

$$H_1 = \frac{S}{g^2}, \qquad H_2 = \frac{T}{g}.$$

Because A has no terms below degree two and C has no constant term,

$$H_1 = \sum_{k \geq 2} A_k u^k g^{k-2} + u, \qquad H_2 = \sum_{k \geq 1} C_k u^k g^{k-1} + 2,$$

so $H_1, H_2 \in \mathbb{C}[s,t]$. A monomial $s^i t^j$ in $H_m(x^2 z, xy)/x^m$ produces $x^{2i+j-m} z^i y^j$. Thus polynomiality is equivalent to the absence of weight less than 2 in H_1 and weight less than 1 in H_2. The only possible offenders are the constant terms of H_1, H_2 and the coefficient of t in H_1. At $(s,t) = (0,0)$ these vanish respectively exactly when (2), (3), and, after differentiating $A((1+t)h(t))/h(t)^2 + 1 + t$, (4) hold.

Conversely, if (1) is polynomial, $H_1(0,0)$ and $H_2(0,0)$ must be defined. If $a = 0$, use $g = h(t) - s$ and evaluate along $t = 0$. The quotient $S/g^2 = A(g)/g^2 + 1$ is defined, but $T/g = C(g)/g + 2$ is also defined; polynomiality after division by x^2 and x forces their constant terms at $s = 0$ to vanish. In terms of $e_0 = E(0)$ these are $e_0/2 + 1 = 0$ and $2e_0 + 2 = 0$, which demand $e_0 = -2$ and $e_0 = -1$, an impossibility. Hence $a \neq 0$, and the preceding weight argument gives all three conditions. This also explains the degree-zero case: when $E = e$ is constant and $a \neq 0$, (2) gives $e = -2$ while (3) gives $e = -1$. The constraints are inconsistent; they do not "force $a = 0$."

For the Jacobian, introduce $w = 1/u$. In coordinates (ν, w),

$$S = A(\nu) + \nu^2 w, \qquad T = C(\nu) + 2\nu w.$$

Using $A' = \nu E$ and $C' = 2E$ gives

$$\frac{\partial(S,T)}{\partial(\nu,w)} = 2\nu^2 w, \qquad \frac{\partial(\nu,w)}{\partial(s,t)} = u^{-1}, \qquad \frac{\partial(S,T)}{\partial(s,t)} = 2g^2.$$

On $x \neq 0$, compare the volume forms in coordinates (x,s,t) and use the third target coordinate xg; the $x\,dg$ term disappears when wedged with $ds \wedge dt$. One obtains

$$F^*(dw_1 \wedge dw_2 \wedge dw_3) = -2\,dx \wedge dy \wedge dz.$$

As F is polynomial, the identity holds everywhere.

It remains to determine the degree. The change

$$(s,t) \dashrightarrow (\nu, w) = ((1+t)(h(t) - s), (1+t)^{-1})$$

is birational, with $t = w^{-1} - 1$ and $s = h(w^{-1} - 1) - \nu w$. Eliminating w from $S = \alpha$ and $T = \beta$ gives

$$A(\nu) - \frac{1}{2}\nu C(\nu) + \frac{\beta}{2}\nu = \alpha. \tag{5}$$

If e_n is the leading coefficient of E, then the coefficient of ν^{n+2} in $2A - \nu C$ is

$$2e_n\left(\frac{1}{n+2} - \frac{1}{n+1}\right) = -\frac{2e_n}{(n+1)(n+2)} \neq 0.$$

Hence (5) has degree exactly $n+2$ and has $n+2$ simple roots generically. Each root reconstructs a unique (s,t) in the birational chart, and over a target with third coordinate $c \neq 0$ one then uniquely recovers $x = c/g$, $y = t/x$, and $z = s/x^2$. Therefore the geometric degree is exactly $n+2$. □

3. Existence in every degree at least three

Let L_n be the Legendre polynomial normalized by $L_n(1) = 1$.

Lemma 3.1 (Shifted Legendre moments)**.** *For $n \geq 1$,*

$$\int_0^1 L_n(2\nu - 1)\, d\nu = 0,$$

and for $n \geq 2$,

$$\int_0^1 \nu L_n(2\nu - 1)\, d\nu = 0.$$

Moreover $L_n(1) = 1$.

Proof. After $r = 2\nu - 1$, the first integral is one half of $\int_{-1}^1 L_n(r)\, dr$, which vanishes by orthogonality to the constant polynomial for $n \geq 1$. The second is one quarter of $\int_{-1}^1 (r+1)L_n(r)\, dr$, which vanishes for $n \geq 2$ by orthogonality to every polynomial of degree at most one. The endpoint normalization is part of the standard definition (and follows immediately from Rodrigues' formula). □

Theorem 3.2 (Universal existence)**.** *For every $n \geq 1$ there are E of degree n and h satisfying Theorem 2.1. Consequently every integer $d \geq 3$ is the geometric degree of a Keller self-map of $\mathbb{A}^3_{\mathbb{C}}$ with determinant -2.*

Proof. For $n = 1$, take

$$E(\nu) = 2 - 6\nu, \qquad h(t) = 1 - \frac{3}{2}t.$$

Then $A(1) = -1$, $C(1) = -2$, $E(1) = -4$, and $1 - 4 + (-3/2)(-2) = 0$.

For $n \geq 2$, take

$$E(\nu) = 2 - 6\nu + L_n(2\nu - 1), \qquad h(t) = 1 - 2t. \tag{6}$$

The two identities of Lemma 3.1 give

$$C(1) = 2\int_0^1 E = -2, \qquad A(1) = \int_0^1 \nu E = -1.$$

Also

$$E(1) = 2 - 6 + L_n(1) = -3;$$

hence the third condition is $1 - 3 + (-2)(-1) = 0$. In all cases $a = h(0) = 1 \neq 0$, so Theorem 2.1 applies and gives degree $n+2$. □

4. An explicit quintic with geometric monodromy S_5

Take $E = 4 - 15\nu + 10\nu^3$ and $h = 1$. Thus

$$A(\nu) = 2\nu^2 - 5\nu^3 + 2\nu^5, \quad C(\nu) = 8\nu - 15\nu^2 + 5\nu^4, \quad g = 1 - x^2 z, \quad \nu = (1+xy)g.$$

The map is compactly

$$F_5 = \left(\frac{A(\nu) + (1+xy)g^2}{x^2 g^2}, \frac{C(\nu) + 2g}{xg}, xg \right). \tag{7}$$

Theorem 2.1 proves polynomiality and gives $\det \mathrm{Jac}(F_5) = -2$ and $\deg_{\mathrm{geom}} F_5 = 5$.

Let (a, b, c) denote target coordinates. Since $c = xg$, the fiber equations imply

$$ac^2 = A(\nu) + \nu g, \qquad bc = C(\nu) + 2g.$$

Eliminating g gives the primitive equation

$$P_{a,b,c}(\nu) = \nu^5 - 5\nu^3 + 4\nu^2 - bc\nu + 2ac^2 = 0. \tag{8}$$

Conversely, away from the proper loci where a denominator vanishes, a root reconstructs the source point by

$$g = \frac{bc - C(\nu)}{2}, \quad x = \frac{c}{g}, \quad u = \frac{\nu}{g}, \quad y = \frac{u-1}{x}, \quad z = \frac{1-g}{x^2}. \tag{9}$$

These are exact rational identities modulo (8).

Theorem 4.1. *The arithmetic and geometric monodromy groups of F_5 are S_5. Consequently the generic algebraic inverse is not solvable by radicals over $\mathbb{C}(a, b, c)$.*

Proof. At the targets

$$(2/3, -1/5, 3/4), \quad (1/2, 1/3, -2/7), \quad (-3/5, 4/9, 1/6),$$

the primitive integral specializations of (8) are

$$\begin{aligned} Q_1 &= 20\nu^5 - 100\nu^3 + 80\nu^2 + 3\nu + 15, \\ Q_2 &= 147\nu^5 - 735\nu^3 + 588\nu^2 + 14\nu + 12, \\ Q_3 &= 270\nu^5 - 1350\nu^3 + 1080\nu^2 - 20\nu - 9. \end{aligned}$$

Exact finite-field factorizations include

$$\begin{aligned} Q_1 \bmod 7 &= 6(\nu^5 + 2\nu^3 - 3\nu^2 - 3\nu - 1), && \text{irreducible}, \\ Q_1 \bmod 103 &= 20(\nu + 11)(\nu + 18)(\nu + 21)(\nu^2 - 50\nu + 40). \end{aligned}$$

The primes are unramified. Thus the arithmetic group contains a 5-cycle and a transposition; transitivity and prime degree imply primitivity, and a primitive subgroup of S_5 containing a transposition is S_5.

For geometric monodromy, (8) is irreducible over $\mathbb{C}(a, b, c)$. Indeed, over $K = \mathbb{C}(b, c)$ it is monic and primitive in $K[a][\nu]$, and it has degree one in a with nonzero coefficient $2c^2 \in K^*$. In a factorization, one factor has a-degree zero; comparison of the coefficient of a makes that factor divide a unit of K, so it is a unit. Hence the geometric group is transitive and normal in the arithmetic S_5, leaving A_5 or S_5.

The three exact specialized discriminants are

$$\begin{aligned} \Delta_1 &= 797038565664000, \\ \Delta_2 &= -244036829099277031104, \\ \Delta_3 &= 1712571399306289125000. \end{aligned}$$

Neither $\Delta_1\Delta_2$ nor $\Delta_1\Delta_3$ is a rational square. If the geometric group lay in A_5, the generic discriminant over $\mathbb{Q}(a,b,c)$ would be a constant times a rational square, forcing every product of two good rational specializations to be a rational square. This contradiction gives geometric group S_5.

Finally, a radical expression for the generic algebraic inverse would put the extension generated through (9) inside a radical tower. Its Galois closure would then have solvable group, contrary to S_5. □

Remark 4.2. The last conclusion concerns the generic algebraic inverse. It does not deny local holomorphic inverses: since $\det \mathrm{Jac}(F_5) = -2$, the holomorphic inverse function theorem supplies a holomorphic inverse germ at every point.

5. The degree-four member

For $E = 3 - 12\nu + 6\nu^2$ and $h = 1 - 2t$, Theorem 2.1 gives a degree-four Keller map F_4. Its primitive generic-fiber equation is

$$\nu^4 - 4\nu^3 + 3\nu^2 - bc\nu + 2ac^2 = 0. \tag{10}$$

Proposition 5.1. *The arithmetic and geometric monodromy groups of F_4 are S_4.*

Proof. At $(a,b,c) = (2/3, -1/5, 3/4)$ the primitive polynomial is

$$R = 20\nu^4 - 80\nu^3 + 60\nu^2 + 3\nu + 15.$$

Its exact unramified factorizations have types (4) modulo 11, $(1,3)$ modulo 7, and $(1,1,2)$ modulo 19. Thus the arithmetic group is transitive and contains a 4-cycle, a 3-cycle, and a transposition; it is S_4.

The geometric group is normal and transitive in S_4, so it is V_4, A_4, or S_4. In the V_4 case the degree-four extension would itself be Galois (the transitive V_4 action is regular). The classical Galois case of the Jacobian problem [1] says that a Keller map whose induced function-field extension is Galois is invertible, contradicting degree four. To exclude A_4, use the three specializations above; their discriminants are

$$-54873514800, \quad -466029671480352, \quad 4549902124608000.$$

The product of the first and third is negative and hence not a rational square. The same constant-times-square specialization argument used for S_5 excludes A_4. Therefore the geometric group is S_4. □

6. Degree spectrum

We take $\mathbb{N} = \{1, 2, 3, \ldots\}$.

Corollary 6.1. *For every $N \geq 3$, the geometric-degree spectrum of Keller self-maps of $\mathbb{A}^N_{\mathbb{C}}$ is*

$$\mathbb{N} \setminus \{2\} = \{1, 3, 4, 5, \ldots\}.$$

Proof. Degree one is realized by automorphisms. Every degree $d \geq 3$ occurs on $\mathbb{A}^3$ by Theorem 3.2, and adjoining $N-3$ identity coordinates preserves the degree and the Keller property.

For completeness, degree two cannot occur. A separable extension of degree two is Galois, and the classical Galois case [1] then makes the Keller map invertible, hence of degree one. This degree-two exclusion is an already-known corollary of the classical theorem and is not claimed as new. □

7. Reproducibility

The file `verify.py` uses Python 3 and SymPy exact arithmetic. It verifies the shifted-Legendre constraints for a range of indices, the polynomiality and Jacobian identities for representative family members, the quintic and quartic primitive equations and rational reconstruction, all stated finite-field certificates, and all specialized discriminants. A successful run ends with `ALL CHECKS PASSED`. No numerical root counts are used.

References

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